Chapter 5 Dimensional Analysis and Similarity

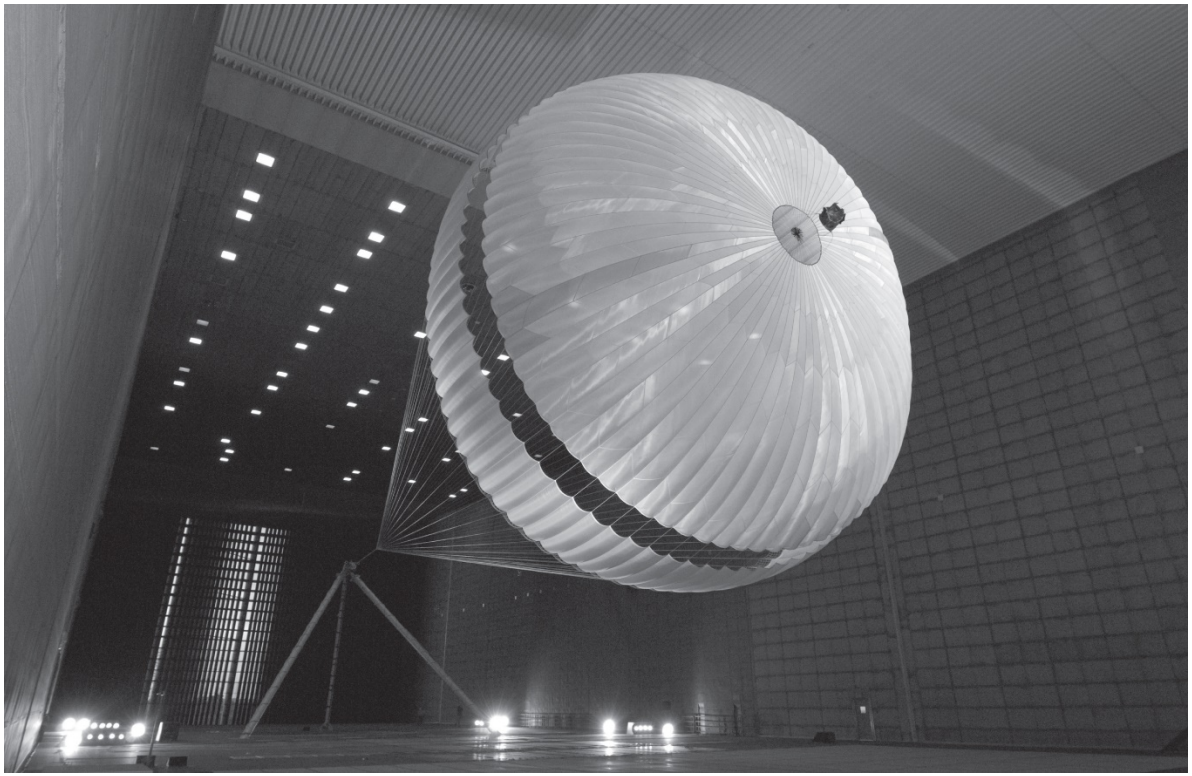


Image from JPL-Caltech/NASA.

Learning Objectives

- Discuss the planning, presentation, and interpretation of experimental data.
- Introduce the **Pi theorem** as the fundamental to the dimensional analysis.
- Present the step-by-step procedure to find out the nondimensional numbers for a complex phenomena.

- Introduce important dimensionless parameters which are widely used in thermal-fluids engineering.
- Introduce a technique for nondimensionalization of the basic equations.
- Understand the modeling process for the flow conditions in engineering applications.

- Analyze similarity in particular types such as *geometric, kinematic, dynamic, and thermal*.

Introduction

- The motivations are two folds:
  1. Dimensional analysis to make the experimental approach to solve a complex fluid mechanics problem more efficient and effective.
  2. Modeling and similitude to justify a model test in lab can be used to describe the behavior of other similar systems outside the lab.





출처 https://www.blogbeforeflight.net/2021/04/south-korea-rolls-out-new-kai-kf-21.html
항공우주연구원 비디오 V_A_ETC_160704_0002

Introduction

- Suppose one knew that the force F on a particular body shape immersed in a stream of fluid depends only on the body length L, stream velocity V, fluid density ρ, and viscosity μ; that is,

$$F = f\left(L, V, \rho, \mu\right)$$

How could the experiments be planned and implemented?

- With the dimensional analysis, we can immediately reduce the above relation to the equivalent form

$$\frac{F}{\rho V^2 L^2} = g\left(\frac{\rho V L}{\mu}\right) \quad \text{or} \quad C_F = g\left(\mathrm{Re}\right)$$

- The **scaling law** states that if $\mathrm{Re}_m \equiv \mathrm{Re}_p$ then $C_{Fm} \equiv C_{Fp}$

The Principle of Dimensional Homogeneity

The *principle of dimensional homogeneity* (PDH) states that for a proper relation, each of its additive items will have the same dimensions.

For example, the Bernoulli equation: $\frac{p}{\rho}+\frac{1}{2}V^2+gz=\text{const}$

Check each term to verify this using MLTΘ dimension:

$$\{p\}=\{ML^{-1}T^{-2}\}\quad\{\rho\}=\{ML^{-3}\}\quad\{V\}=\{LT^{-1}\}\quad\{g\}=\{LT^{-2}\}\quad\{z\}=\{L\}$$

So each term has a dimension of $\{L^2T^{-2}\}$. Dimensions of fluid mechanics properties can refer to Table 5.1.

The Pi Theorem

Variables and constants: for example, $\frac{p}{\rho}+\frac{1}{2}V^2+gz=\text{const}$

- **Dimensional variables** are the quantities varying in a given case and would be plotted against each other to show the data. (E.g., p, V, and z)
- **Dimensional constants** may vary from case to case but are held constant during a given run. (E.g., ρ and g)
- **Pure constants** have no dimensions and never did. (E.g., π, or e)
- **Angles and revolutions** are dimensionless.

The Pi Theorem

- **The Buckingham Pi Theorem** is one of the methods to reduce a number of dimensional variables into a smaller number of dimensionless groups.

- **How many *dimensionless variables* (Πs) can be formed?**

  If a physical process satisfies the PDH and involves ***n* dimensional variables**, it can be reduced to a relation between ***k* dimensionless variables** or **Π**s

- The reduction $j = n - k$ equals the maximum number of variables that do not form a pi among themselves and is always less than or equal to the number of dimensions describing the variables.

The Pi Theorem

Example: Consider force on an immerse body $F = f\left(L, V, \rho, \mu\right)$, how many Pi groups can it be formed?

Solution: The equation contains five variables $F, L, V, \rho,$ and μ, described by three basic dimensions $\{MLT\}$. Thus $n = 5$ and $j \le 3$. Therefore, it is a good guess that we can reduce the problem to k pi groups, with

$k = n - j \ge 5 - 3 = 2.$

This is exactly what we obtained (details will be shown later):

$\Pi_1 = C_F$ and $\Pi_2 = \mathrm{Re}.$

Only on rare occasion it may take more Pi groups than this minimum.

How to Form Pi Groups?

Example: Consider force on an immerse body in fluid, $F = f(L, V, \rho, \mu)$, rewrite the function in terms of pi groups?

Solution:

Step 1: Write the function with all the variables that are involved in the problem:

$F = f(L, V, \rho, \mu)$, there are five variables $(n = 5)$

This step is the most difficult one in real engineering practice.

Ambiguity 1: Which variable should be included or not be included?

Guideline for selecting variables: (1) geometric variables of the system, such as the length of a body, the diameter of a pipe; (2) variables of fluid properties like density, viscosity; (3) variables representing external effects, such as drag force of an immersed body, pressure drop along a pipeline.

How to Form Pi Groups?

Step 2: List dimensions of each variable.

F	L	V	ρ	μ
$\{MLT^{-2}\}$	$\{L\}$	$\{\mathrm{LT}^{-1}\}$	$\{\mathrm{ML}^{-3}\}$	$\{\mathrm{ML}^{-1}\mathrm{T}^{-1}\}$

Step 3: Find the reduction j and determine the number of pi groups to be formed.

The equation contains five variables F, L, V, ρ and μ described by three basic dimensions $\{MLT\}$. Thus $n = 5$ and $j \leq 3$.

Then, we can reduce the problem to k pi groups, with $k = n - j \geq 5 - 3 = 2$.

How to Form Pi Groups?

Step 4: Select *j* scaling (repeating) variables that do not form a pi product.

Rule: Each selected scaling variable must be dimensionally independent of the others. This means that they must not form a dimensionless group among themselves. For example, test powers of L, V,

$$L^a V^b \rho^c = \{L\}^a \{LT^{-1}\}^b \{ML^{-3}\}^c = M^0 L^0 T^0 \quad \text{only if } a = 0, b = 0, c = 0$$

However, if L,V, and μ or $L, \rho,$ and μ are selected, they still obey the rule.

Ambiguity 2: Selection of the scaling variables is not unique.

Guideline for selecting scaling variables: (1) Do not select output variables for your scaling variables; (2) If convenient, select simple and popular, not obscure, scaling variables because they will appear in all of your dimensionless groups.

According to the rule and guideline, we select L, V, ρ as the scaling variables.

How to Form Pi Groups?

Step 5: Combine L, V, ρ with one additional variable to find the two pi products.

First add force F to find Π_1.

Ambiguity 3: You may select any nonzero exponent on this additional term to place it in the numerator or denominator to any power.

We select F to appear to the first power in the numerator.

$$\Pi_1 = L^a V^b \rho^c F = \{L\}^a \{LT^{-1}\}^b \{ML^{-3}\}^c \{MLT^{-2}\} = M^0 L^0 T^0$$

Equate exponents:

Length: $a + b - 3c = 0$

Mass: $c + 1 = 0$

Time: $-b - 2 = 0$

$\Rightarrow$ $a = -2, b = -2, c = -1$

Therefore $\Pi_1 = L^{-2} V^{-2} \rho^{-1} F = \dfrac{F}{\rho V^2 L^2} = C_F$

How to Form Pi Groups?

Next add viscosity μ to find Π_2.

We select the power -1 for μ and place it in the denominator:

$$\Pi_2 = L^a V^b \rho^c \mu^{-1} = \{L\}^a \{LT^{-1}\}^b \{ML^{-3}\}^c \{ML^{-1}T^{-1}\}^{-1} = M^0 L^0 T^0$$

Equate exponents

$$\text{Length: } a + b - 3c + 1 = 0$$
$$\text{Mass: } \quad c - 1 = 0 \quad \Rightarrow \quad a = b = c = 1$$
$$\text{Time: } \quad -b + 1 = 0$$

Therefore $\Pi_2 = L^1 V^1 \rho^1 \mu^{-1} = \dfrac{\rho V L}{\mu} = \text{Re}$

Step 6: Express the final form as a relationship among the pi groups.

$$\frac{F}{\rho V^2 L^2} = g\left(\frac{\rho V L}{\mu}\right)$$

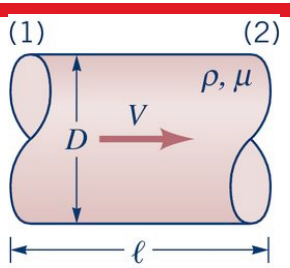
Step 7: Check the dimensions of the pi groups to ensure they are indeed dimensionless.

How to Form Pi Groups?

- Consider the pipe flow problem for an example.

- Let's follow the steps below.

1. List all the necessary & independent ***n*** variables.
2. Express each of the variables in terms of ***j*** basic reference dimensions.
3. Determine the required number of pi terms: ***k=n-j***.
4. Select a number of repeating variables, where the number required is equal to the number of reference dimensions: ***j***.
5. Form a pi term multiplying one of the nonrepeating variables by the product of the repeating variables, each raised to an exponent that will make the combination dimensionless.
6. Repeat Step 5 for each of the remaining nonrepeating variables.
7. Check all the resulting pi terms to make sure they are dimensionless.
8. Express the final form as a relationship among the pi terms, and think about what it means.



$\Delta p_\ell = (p_1 - p_2)/\ell$

$\Delta p_l = f(D, \rho, \mu, V)$

$n = 5;\ D, \rho, \mu, V, \Delta p_l$

Variables / Dimensions	D	ρ	μ	V	Δp_l
M	0	1	1	0	1
L	1	-3	-1	1	-2
T	0	0	-1	-1	-2

rank of dimensional matrix determines ***j***

$j = 3$

$k = n - j = 2$

$D,\ V,\ \rho$

$$\Pi_1 = \mu D^a V^b \rho^c \doteq M^{(1+c)} L^{(-1+a+b-3c)} T^{(-1-b)} = M^0 L^0 T^0$$

$$\therefore\ a = -1, b = -1, c = -1$$

$$\therefore\ \Pi_1 = \frac{\mu}{\rho V D}$$

$$\Pi_2 = \Delta p_l D^a V^b \rho^c \doteq M^{(1+c)} L^{(-2+a+b-3c)} T^{(-2-b)} = M^0 L^0 T^0$$

$$\therefore\ a = 1, b = -2, c = -1$$

$$\therefore\ \Pi_2 = \frac{D\,\Delta p_l}{\rho V^2}$$

$$\Pi_2 = \frac{D\,\Delta p_l}{\rho V^2} = \tilde{\phi}\left(\Pi_1 = \frac{\mu}{\rho V D}\right) \text{ or } \Pi_2 = \frac{D\,\Delta p_l}{\rho V^2} = \phi\left(\frac{\rho V D}{\mu}\right)$$

Dimensionless Parameters

- Using the method of the Pi theorem, the resultant number of dimensionless parameters is fixed.
- However, there is no unique set of dimensionless groups which arises from a dimensional analysis. What are the appropriate Pi groups?

Dimensionless Parameters (Table 5.2)

Parameter	Definition	Qualitative ratio of effects	Importance
Reynolds number	$\mathrm{Re} = \frac{\rho U L}{\mu}$	$\frac{\text{Inertia}}{\text{Viscosity}}$	Almost always
Mach number	$\mathrm{Ma} = \frac{U}{a}$	$\frac{\text{Flow speed}}{\text{Sound speed}}$	Compressible flow
Froude number	$\mathrm{Fr} = \frac{U^2}{gL}$	$\frac{\text{Inertia}}{\text{Gravity}}$	Free-surface flow
Weber number	$\mathrm{We} = \frac{\rho U^2 L}{\Upsilon}$	$\frac{\text{Inertia}}{\text{Surface tension}}$	Free-surface flow
Rossby number	$\mathrm{Ro} = \frac{U}{\Omega_{\text{earth}} L}$	$\frac{\text{Flow velocity}}{\text{Coriolis effect}}$	Geophysical flows
Cavitation number (Euler number)	$\mathrm{Ca} = \frac{p - p_v}{\frac{1}{2}\rho U^2}$	$\frac{\text{Pressure}}{\text{Inertia}}$	Cavitation
Prandtl number	$\mathrm{Pr} = \frac{\mu c_p}{k}$	$\frac{\text{Dissipation}}{\text{Conduction}}$	Heat convection
Eckert number	$\mathrm{Ec} = \frac{U^2}{c_p T_0}$	$\frac{\text{Kinetic energy}}{\text{Enthalpy}}$	Dissipation
Specific-heat ratio	$k = \frac{c_p}{c_v}$	$\frac{\text{Enthalpy}}{\text{Internal energy}}$	Compressible flow
Strouhal number	$\mathrm{St} = \frac{\omega L}{U}$	$\frac{\text{Oscillation}}{\text{Mean speed}}$	Oscillating flow

Dimensionless Parameters (Table 5.2)

- Viscous flows – **Reynolds number**.

$$\mathrm{Re} = \frac{\text{Inertial force}}{\text{Viscous force}} \sim \frac{\rho L^2 V^2}{\mu L V} = \frac{\rho L V}{\mu}$$

- Free-surface flow – **Weber number**.

$$\mathrm{We} = \frac{\text{Inertial force}}{\text{Surface tension force}} = \frac{\rho L^2 V^2}{Y L} = \frac{\rho L V^2}{Y}$$

- Open channel flow – **Froude number**.

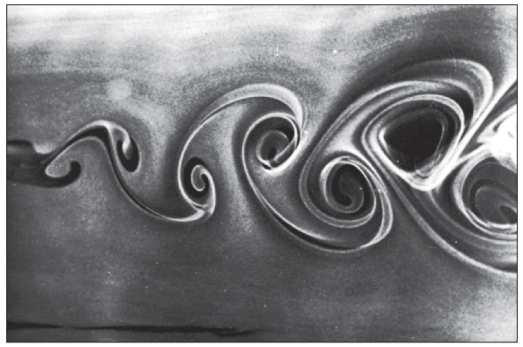
$$\mathrm{Fr} = \frac{\text{Inertial force}}{\text{Gravity fouce}} \sim \frac{\rho L^2 V^2}{\rho L^3 g} = \frac{V^2}{gL} \quad \text{or} \quad \mathrm{Fr} = \frac{\text{Flow speed}}{\text{Wave speed}} \sim \frac{V}{\sqrt{gL}}$$

- Compressible flow – **Mach number**.

$$\mathrm{Ma} = \frac{\text{Flow speed}}{\text{Sound speed}} \sim \frac{V}{a}$$

- Oscillation flow – **Strouhal number**.

$$\mathrm{St} = \frac{\text{Oscillation}}{\text{Mean speed}} \sim \frac{\omega L}{V}$$



(a)

Vortex shedding from a circular cylinder

Dimensionless Parameters (Table 5.2)

Parameter	Definition	Qualitative ratio of effects	Importance
Roughness ratio	$\frac{\varepsilon}{L}$	$\frac{\text{Wall roughness}}{\text{Body length}}$	Turbulent, rough walls
Grashof number	$\text{Gr} = \frac{\beta \Delta T g L^3 \rho^2}{\mu^2}$	$\frac{\text{Buoyancy}}{\text{Viscosity}}$	Natural convection
Rayleigh number	$\text{Ra} = \frac{\beta \Delta T g L^3 \rho^2 c_p}{\mu k}$	$\frac{\text{Buoyancy}}{\text{Viscosity}}$	Natural convection
Temperature ratio	$\frac{T_w}{T_0}$	$\frac{\text{Wall temperature}}{\text{Stream temperature}}$	Heat transfer
Pressure coefficient	$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho U^2}$	$\frac{\text{Static pressure}}{\text{Dynamic pressure}}$	Aerodynamics, hydrodynamics
Lift coefficient	$C_L = \frac{L}{\frac{1}{2}\rho U^2 A}$	$\frac{\text{Lift force}}{\text{Dynamic force}}$	Aerodynamics, hydrodynamics
Drag coefficient	$C_D = \frac{D}{\frac{1}{2}\rho U^2 A}$	$\frac{\text{Drag force}}{\text{Dynamic force}}$	Aerodynamics, hydrodynamics
Friction factor	$f = \frac{h_f}{(V^2/2g)(L/d)}$	$\frac{\text{Friction head loss}}{\text{Velocity head}}$	Pipe flow
Skin friction coefficient	$c_f = \frac{\tau_{\text{wall}}}{\rho V^2/2}$	$\frac{\text{Wall shear stress}}{\text{Dynamic pressure}}$	Boundary layer flow

Reynolds Number $\mathrm{Re} = \frac{\rho V l}{\mu} = \frac{V l}{\nu}$

- Physical meaning: ratio of the inertia force to viscous force

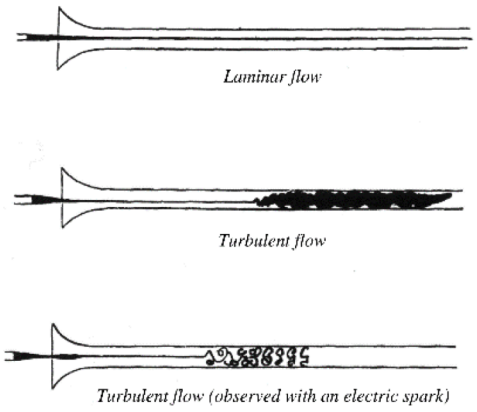
- Inertia force: $\rho \frac{\partial \bar{V}}{\partial t} \doteq \rho \frac{V}{l/V} = \frac{\rho V^2}{l}$

- Viscous force: $\mu \nabla^2 \bar{V} \doteq \frac{\mu V}{l^2}$

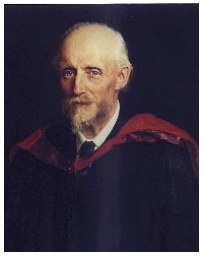
- Ratio: $\frac{\rho V^2}{l} / \frac{\mu V}{l^2} = \frac{\rho V l}{\mu} \rightarrow \mathrm{Re} = \frac{\rho V l}{\mu}$

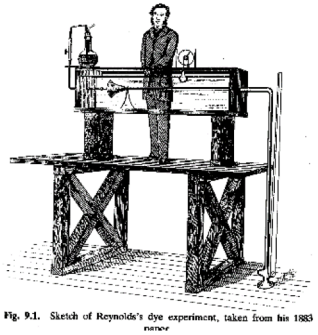
- Fluid flows are commonly categorized with *Re*

- For flow with Re<<1 ("creeping flow"), viscous effects are dominant.
- For flow with Re>>1, we may be able to assume inviscid flow.
- For small Re, flow is usually laminar flow.
- For large Re, flow can be turbulent.



Osborne Reynolds (1842-1912), a British engineer

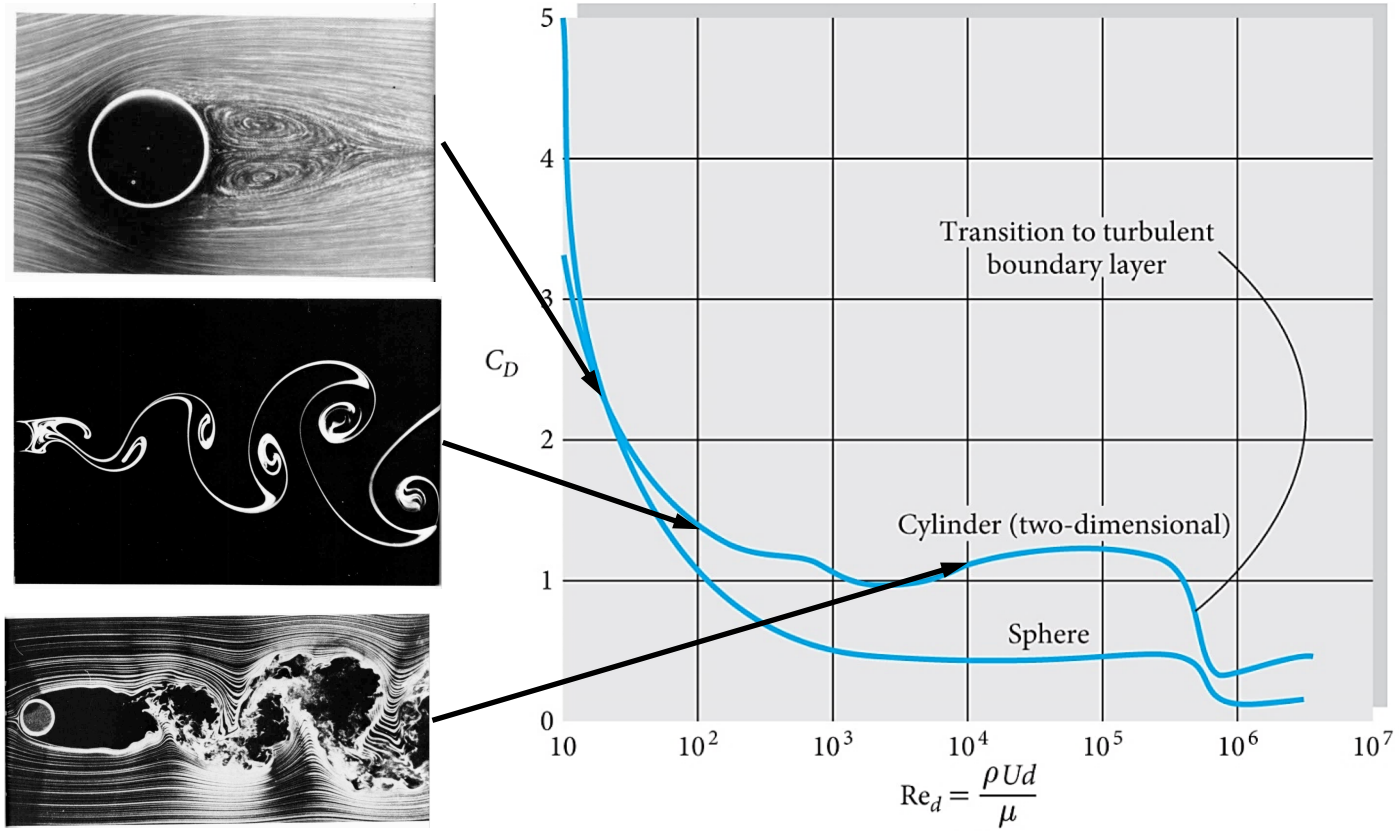



Fig. 9.1. Sketch of Reynolds's dye experiment, taken from his 1883 paper

출처 https://en.wikipedia.org/wiki/Osborne_Reynolds

Drag Coefficient

Drag coefficient (non-dimensional drag force) depends strongly on Reynolds number



출처 Frank White, Fluid Mechanics, 8th Ed., McGraw-Hill Education (Asia), 2016
Images from Van Dyke's "An Album of Fluid Motion"

Mach Number $\mathrm{Ma} = \frac{V}{a}$

- Physical meaning: ration of inertial force to compressibility force

- Inertia force: $\rho \frac{\partial \bar{V}}{\partial t} \doteq \rho \frac{V}{l/V} = \frac{\rho V^2}{l}$

- Compressibility force:

$dp = \beta(d\rho/\rho) = a^2\rho(d\rho/\rho) = a^2 d\rho$ (pp. 34-35 in textbook)

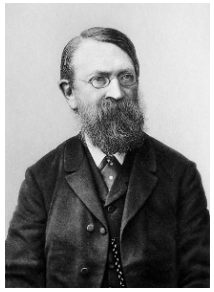
Isentropic bulk modulus $\beta = \rho\left(\frac{\partial p}{\partial \rho}\right)_s$

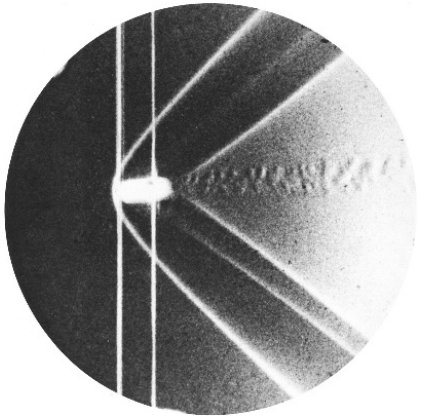
$$\nabla p \doteq \frac{\rho a^2}{l}$$

- Ratio: $\frac{\rho V^2}{l} / \frac{\rho a^2}{l} = \frac{V^2}{a^2} = \frac{\rho V^2}{\rho a^2} = \left(\frac{V}{a}\right)^2 = \mathrm{Ma}^2$

Ernst Mach (1838-1916), an Austrian physicist

출처 https://en.wikipedia.org/wiki/Ernst_Mach





Shadowgraph of a bow shockwave around a supersonic bullet

Strouhal Number $St = \frac{\omega l}{V}$

- Physical meaning: ratio of inertia force (local acceleration due to unsteady oscillation) to inertia force (convective acceleration)

  - Local acceleration: Local acceleration: $\rho \frac{\partial V}{\partial t} \doteq \rho \frac{V}{l/V} = \frac{\rho V^2}{l} = \frac{\rho (l\omega)^2}{l} = \rho l \omega^2$

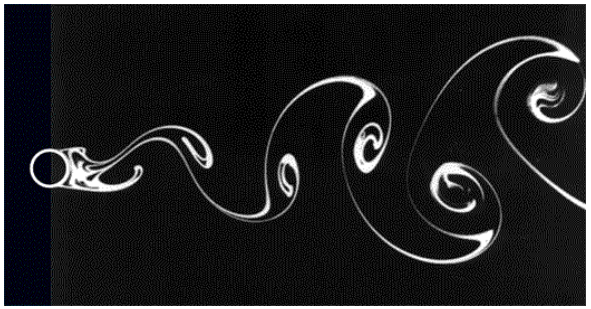
  - Convective acceleration: $\rho V \cdot \nabla V \doteq \frac{\rho V^2}{l}$

  - Ratio: $\rho l \omega^2 / \frac{\rho V^2}{l} = \frac{l^2 \omega^2}{V^2}$

- Flow separation from a body creates an organized pattern of vortices

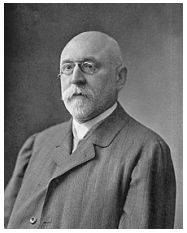
a von Kármán vortex street, with a discrete frequency ω.

- If ω is in the audible range, the flow creates a sound, and the body appears to "sing".



Images from Van Dyke's "An Album of Fluid Motion"

Vincenc Strouhal (Czech physics, 1850-1922) studied "singing wires."



https://en.wikipedia.org/wiki/Vincenc_Strouhal

Nondimensionalization of the Governing Equations

- An alternative and very powerful technique to find dimensionless groups for the governing equations of flow from Chapter 4.

Nondimensionlize the 2-D incompressible flow continuity and momentum equations with constant viscosity.

Continuity:

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}=0$$

Momentum:

$$\rho\left(\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\right)=-\frac{\partial p}{\partial x}+\mu\left(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}\right)$$

$$\rho\left(\frac{\partial v}{\partial t}+u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}\right)=-\frac{\partial p}{\partial y}-\rho g+\mu\left(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}\right)$$

Boundary conditions for these equations are

Fixed solid surface: $u=0, v=0$

Inlet or outlet: given u, v, p

Once the governing equations, including boundary conditions, are known, we are ready to proceed to develop similarity requirements.

Nondimensionalization of the Governing Equations

Equations contain the three basic dimensions, M, L, and T. All the variables p, u, v, x, y, t can be nondimensionalized by using a reference length L, a reference velocity V, and reference pressure p_0 that might be characteristic of the particular fluid flow.

Step 1: Define all relevant dimensionless variables,

$$u^* = \frac{u}{V} \quad v^* = \frac{v}{V} \quad p^* = \frac{p}{p_0} \quad x^* = \frac{x}{L} \quad y^* = \frac{y}{L} \quad t^* = \frac{tV}{L}$$

Ambiguity: You may choose different reference, for example,

$$p^* = \frac{p + \rho g y}{\rho V^2} \quad t^* = \frac{t}{\tau}$$ where τ is a reference time

Step 2: Rewrite the governing equations in terms of new variables, for example,

$$\frac{\partial u}{\partial x} = \frac{\partial \left(u^* V \right)}{\partial \left(x^* L \right)} = \frac{V}{L} \frac{\partial u^*}{\partial x^*}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{V}{L} \frac{\partial}{\partial \left(x^* L \right)} \left(\frac{\partial u^*}{\partial x^*} \right) = \frac{V}{L^2} \frac{\partial^2 u^*}{\partial x^{*2}}$$

Nondimensionalization of the Governing Equations

$$\frac{\partial u^*}{\partial x^*}+\frac{\partial v^*}{\partial y^*}=0$$

$$\left[\frac{\rho V^2}{L}\right]\left(\frac{\partial u^*}{\partial t^*}+u^*\frac{\partial u^*}{\partial x^*}+v^*\frac{\partial u^*}{\partial y^*}\right)=-\left[\frac{p_0}{L}\right]\frac{\partial p^*}{\partial x^*}+\left[\frac{\mu V}{L^2}\right]\left(\frac{\partial^2 u^*}{\partial x^{*2}}+\frac{\partial^2 u^*}{\partial y^{*2}}\right)$$

$$\left[\frac{\rho V^2}{L}\right]\left(\frac{\partial v^*}{\partial t^*}+u^*\frac{\partial v^*}{\partial x^*}+v^*\frac{\partial v^*}{\partial y^*}\right)=-\left[\frac{p_0}{L}\right]\frac{\partial p^*}{\partial y^*}-\rho g+\left[\frac{\mu V}{L^2}\right]\left(\frac{\partial^2 v^*}{\partial x^{*2}}+\frac{\partial^2 v^*}{\partial y^{*2}}\right)$$

At solid surface: $u^*=0 \quad v^*=0$, Inlet or outlet: given u^*, v^*, p^*

Step 3: Divide the dimension term $\rho V^2/L$, and identify the dimensionless groups.

$$\left(\frac{\partial u^*}{\partial t^*}+u^*\frac{\partial u^*}{\partial x^*}+v^*\frac{\partial u^*}{\partial y^*}\right)=-\left[\frac{p_0}{\rho V^2}\right]\frac{\partial p^*}{\partial x^*}+\left[\frac{\mu}{\rho VL}\right]\left(\frac{\partial^2 u^*}{\partial x^{*2}}+\frac{\partial^2 u^*}{\partial y^{*2}}\right)$$

$$\left(\frac{\partial v^*}{\partial t^*}+u^*\frac{\partial v^*}{\partial x^*}+v^*\frac{\partial v^*}{\partial y^*}\right)=-\left[\frac{p_0}{\rho V^2}\right]\frac{\partial p^*}{\partial y^*}-\frac{gL}{V^2}+\left[\frac{\mu}{\rho VL}\right]\left(\frac{\partial^2 v^*}{\partial x^{*2}}+\frac{\partial^2 v^*}{\partial y^{*2}}\right)$$

where $\mathrm{Eu}=\dfrac{p_0}{\rho V^2} \quad \dfrac{1}{\mathrm{Fr}}=\dfrac{gL}{V^2} \quad \dfrac{1}{\mathrm{Re}}=\dfrac{\mu}{\rho VL}$

Modeling and Similarity

Flow conditions for a model test are completely similar if all relevant dimensionless parameters have the same corresponding values for the model and the proto type. Mathematically it implies,

for a desired dimensionless function,

$$\Pi_1 = f\left(\Pi_2, \Pi_3, \cdots \Pi_k\right)$$

If $\Pi_{2m} = \Pi_{2p}, \Pi_{3m} = \Pi_{3p}, \cdots \Pi_{km} = \Pi_{kp}$

then $\Pi_{1m} = \Pi_{1p}$

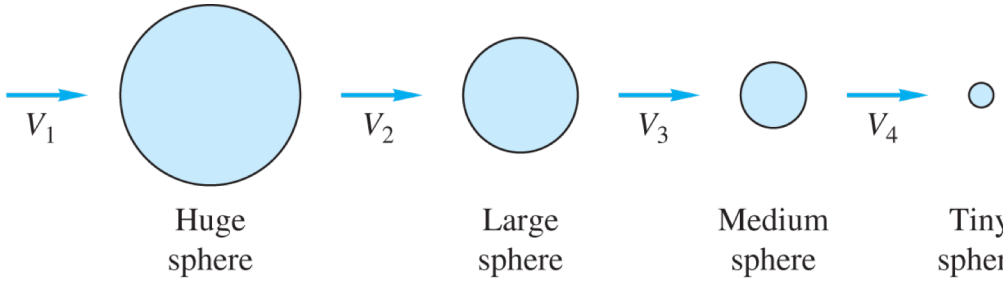
This is straightforward mathematically, but certain engineering difficulties need to be reminded:

- For dimensional analysis, selection of the important variables requires considerable judgment and experience. The engineer must decide, for example, whether viscosity can be neglected? Are there significant temperature effects?
- It is sometimes very difficult to preserve all the pi groups equal between the model and prototype – pitfalls.

Geometric Similarity

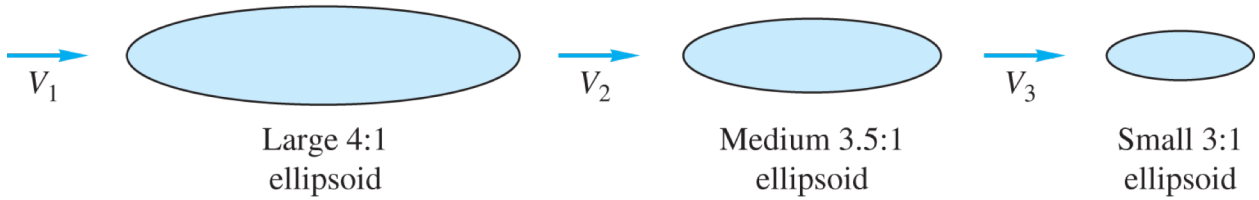
- **Geometric similarity**: A model and prototype are geometrically similar <u>if and only if all body dimensions in all three coordinates have the same *linear* scale ratio</u>.

Similar



(*a*)

Similar?

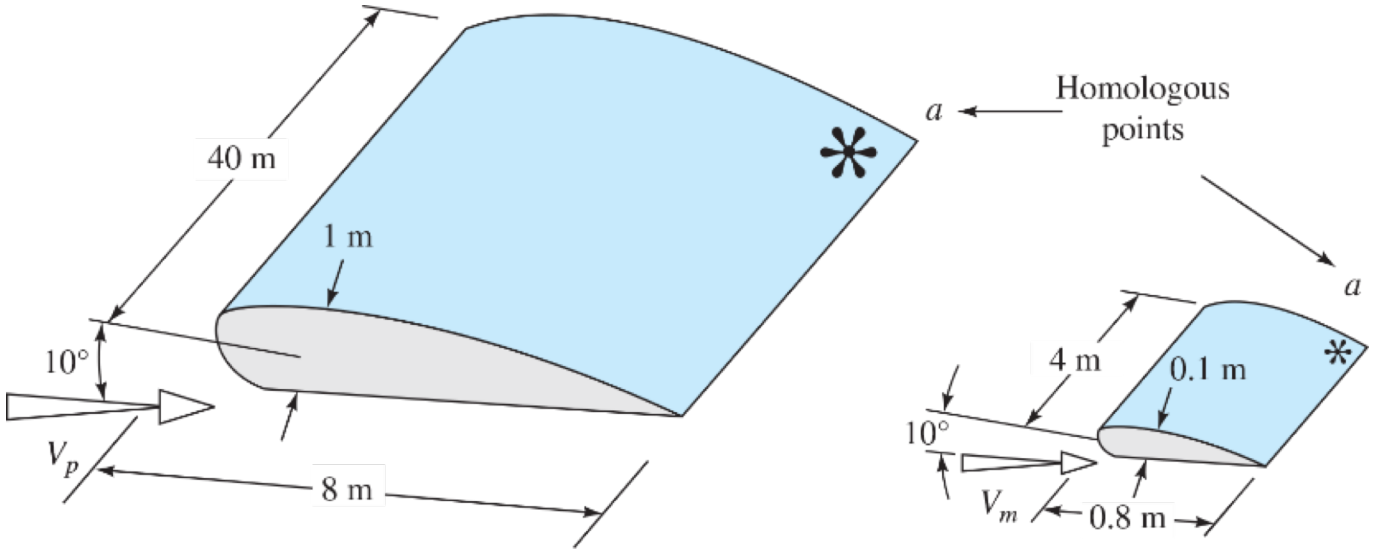


(*b*)

Geometric Similarity

- **Geometric similarity**: A model and prototype are geometrically similar if and only if all body dimensions in all three coordinates have the same *linear* scale ratio.

- All angles and flow directions are preserved. The orientations of model and prototype must be identical.



Full prototype

1/10th scale model

Kinematic Similarity

- **Kinematic similarity**: The motions of two systems are kinematically similar if homologous particles lie at homologous points at homologous times: They have a similar position, structure, value, or origin, even if the function has changed over time

- Kinematic similarity requires that the model and the prototype have the same length scale ratio and the same time scale ratio. Then, the velocity scale ratio will be the same for both.

- Two types of kinematic similarity:

  1. incompressible frictionless flow without free surface

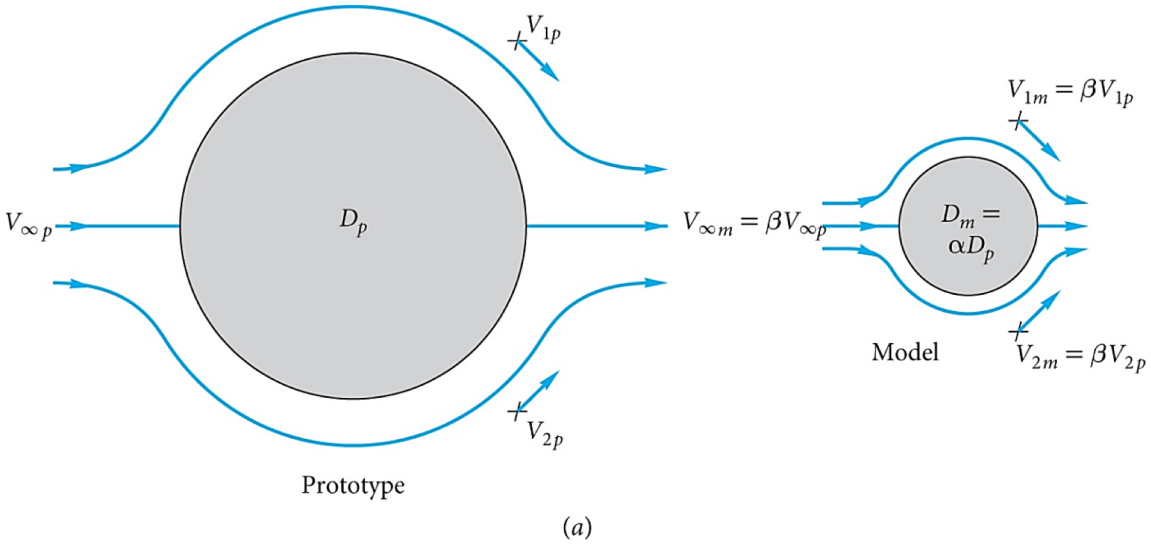
  2. incompressible frictionless flow with free surface but surface tension is not important.

Kinematic Similarity

- Incompressible frictionless flow without free surface,

One special case: in incompressible and frictionless flows with no free surface, flows are similar with independent length scale and time scale.

- E.g., $\frac{V_m}{V_p} = \text{const}$ throughout the flow field

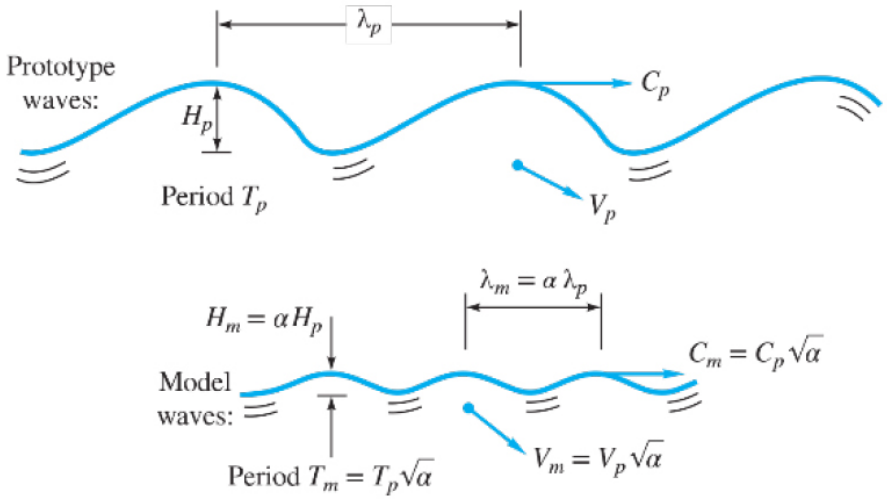


Kinematic Similarity

- Incompressible frictionless flow with free surface but surface tension is not important.

$$\mathrm{Fr} = \frac{\text{Flow speed}}{\text{Wave speed}} \sim \frac{V}{\sqrt{gL}}$$

(Ratio of inertial forces to gravitational forces)



- Froude scaling $\mathrm{Fr_m} = \dfrac{V_m}{\sqrt{gL_m}} = \dfrac{V_p}{\sqrt{gL_p}} = \mathrm{Fr_p}$

If $L_m = \alpha L_p$ Then $\dfrac{V_m}{V_p} = \left(\dfrac{L_m}{L_p}\right)^{1/2} = \sqrt{\alpha}$ $\qquad \dfrac{T_m}{T_p} = \dfrac{L_m/V_m}{L_p/V_p} = \sqrt{\alpha}$

Dynamic Similarity

Dynamic similarity: The model and prototype have the same length scale ratio, time scale ratio, and **force scale** (or mass scale) ratio.

Two types of dynamics similarity:

1. For compressible flow, $\mathrm{Re}_m = \mathrm{Re}_p \quad \mathrm{Ma}_m = \mathrm{Ma}_p \quad k_m = k_p$
2. For incompressible flow,
   - with no free surface $\mathrm{Re}_m = \mathrm{Re}_p$
   - with a free surface $\mathrm{Re}_m = \mathrm{Re}_p \quad \mathrm{Fr}_m = \mathrm{Fr}_p \quad \mathrm{We}_m = \mathrm{We}_p$

Modeling and Similarity

Example: The pressure drop due to friction for flow in a long, smooth pipe is a function of average flow velocity, density, viscosity, and pipe length and diameter, $\Delta p = f(V, \rho, \mu, L, D)$. We wish to know Δp varies with V. (a) Use the pi theorem to write the dimensionless form. (b) Plot the function using the data from three pipes and three fluids below. (c) Assume Δp is proportional to L, simplify and improve the pi theorem formulation.

D, cm	L, m	Q, m^3/h	Δp, Pa	ρ, kg/m^3	μ, $kg/m \cdot s$	V, m/s
1.0 (gasoline)	5.0	0.3	4,068	680	2.92E-4	1.06
1.0 (gasoline)	7.0	0.6	22,300	680	2.92E-4	2.12
1.0 (gasoline)	9.0	1.0	70,800	680	2.92E-4	3.54
2.0 (water)	4.0	1.0	2,080	998	0.0010	0.88
2.0 (water)	6.0	2.0	10,500	998	0.0010	1.77
2.0 (water)	8.0	3.1	30,400	998	0.0010	2.74
3.0 (mercury)	3.0	0.5	540	13,550	1.56E-3	0.20
3.0 (mercury)	4.0	1.0	2,480	13,550	1.56E-3	0.39
3.0 (mercury)	5.0	1.7	9.600	13,550	1.56E-3	0.67

Modeling and Similarity

Solution: There are six variables with three basic dimensions, M, L, and T. So we expect $j = 6 - 3 = 3$ pi groups. We select (ρ, μ, D) as the scaling variables to ensure that the three pi groups will contain Δp, V, and L, and we have to plot Δp or V.

$$\Pi_1 = \rho^a \mu^b D^c \Delta p \qquad \Pi_2 = \rho^d \mu^e D^f V \qquad \Pi_3 = \rho^g \mu^h D^l V$$

$$\Pi_1 = \frac{\rho D^2 \Delta p}{\mu^2} \qquad \Pi_2 = \frac{\rho V D}{\mu} \qquad \Pi_3 = \frac{L}{D}$$

Modeling and Similarity

Solution (Con'd)

Therefore, we wish to plot the dimensionless relation

$$\frac{\rho D^2 \Delta p}{\mu^2} = f\left(\frac{\rho V L}{\mu}, \frac{L}{D}\right)$$

We plot Π_1 versus Π_2 and Π_3 as a parameter. For example, the first row in the data

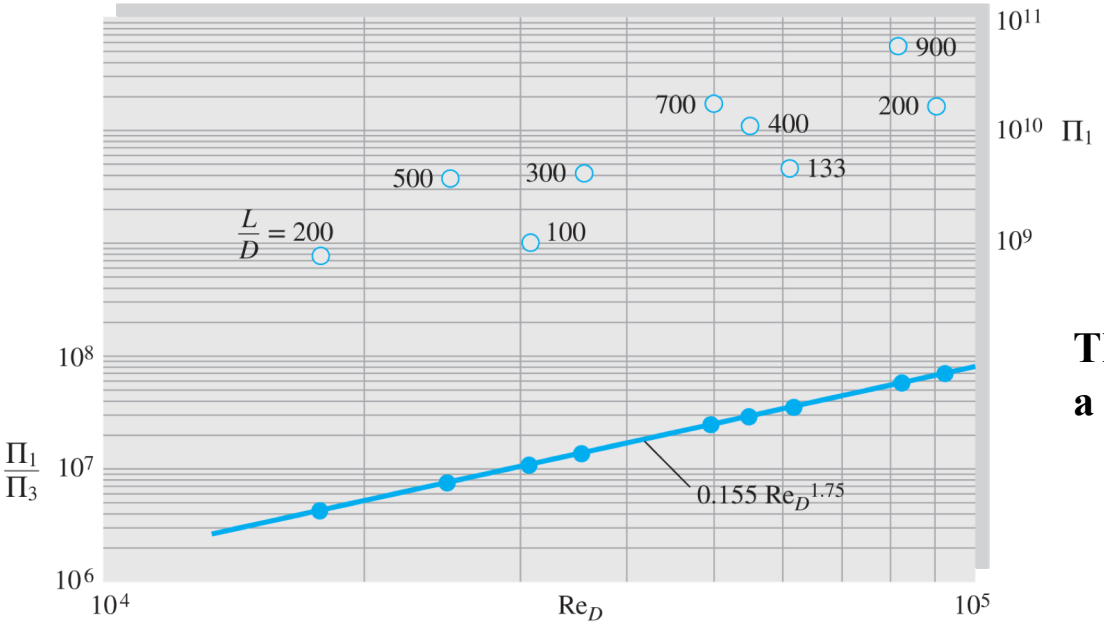
$$\Pi_1 = \frac{\rho D^2 \Delta p}{\mu^2} = \frac{(680)(0.01)^2(4680)}{(2.92\text{E}-4)^2} = 3.73\text{E}9$$

$$\Pi_2 = \frac{\rho V L}{\mu} = \frac{(680)(1.06)(0.01)}{2.92\text{E}-4} = 24,700$$

$$\Pi_3 = \frac{L}{D} = 500$$

The total nine points are plotted as the open circles in figure below. The values of L/D are listed for each point, and we see a significant length effect.

Modeling and Similarity



They correlate almost perfectly into a straight line power-law function

$$\frac{\rho D^3 \Delta p}{L\mu^2} \approx 0.155\left(\frac{\rho VL}{\mu}\right)^{1.75}$$

Since Δp increases linearly with L that occurs only in Π_3, function $\Pi_1 = f\left(\Pi_2, \Pi_3\right)$ can be reduced to $\Pi_1 = \left(L/D\right) f\left(\Pi_2\right)$ or simply as

$$\frac{\rho D^3 \Delta p}{L\mu^2} = f\left(\frac{\rho VL}{\mu}\right)$$

Summary

- The third method for analyzing fluid flow is introduced as experimentation, as supplemented by the technique of dimensional analysis.
- A general technique, the pi theorem, is presented for systematically finding a set of dimensionless parameters by grouping a list of variables that govern any physical process.
- Alternatively, direct application of dimensional analysis to the basic governing equations of fluid mechanics yields the fundamental parameters governing the flow patterns such as Reynolds number, Froude number, Mach number, and others.
- Flow conditions for a model test are completely similar if all relevant dimensionless parameters have the same corresponding values for the model and the proto type.
- Instead of complete similarity, the most common similarity for particular types are geometric similarity, kinematic similarity, and kinetic similarity.
- Classic dimensionless charts and data can be manipulated and recast to provide direct solutions to problems.